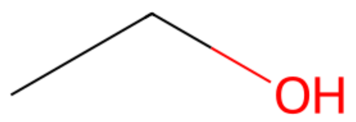
# QSAR Irritation Prediction Report

### Eye & Skin Irritation Toxicity Assessment

## 1. PREDICTION SUMMARY

| | |
|---|---|
| **Compound Name:** | Ethanol |
| **SMILES:** | CCO |
| **Report Date:** | 2026-06-29 17:47:01 |




| Endpoint | Prediction | Probability | Confidence | Threshold |
|---|---|---|---|---|
| Eye Irritation | Non-irritant | 0.250 | High | 0.32 |
| Skin Irritation | Non-irritant | 0.300 | Medium | 0.47 |

**Training Data:**

• Eye Irritation: NTP ICE In Vivo data (EPA classification)

• Skin Irritation: NTP ICE In Vivo data (EPA classification)

**Models:**

• Eye: Gradient Boosting Classifier (ROC-AUC: 0.765, Sensitivity: 81.7%)

• Skin: Support Vector Machine - RBF kernel (ROC-AUC: 0.756, Sensitivity: 76.5%)

## 2. MOLECULAR DESCRIPTORS

| Descriptor | Value |
|---|---|
| MolWt | 46.069 |
| MolLogP | -0.031 |
| TPSA | 20.230 |
| NumHAcceptors | 1 |
| NumHDonors | 1 |
| NumRotatableBonds | 0 |
| NumAromaticRings | 0 |
| FractionCSP3 | 0.667 |
| NumHeteroatoms | 1 |
| NumSaturatedRings | 0 |
| NumAliphaticRings | 0 |
| RingCount | 0 |
| HeavyAtomCount | 3 |
| Mixture | 0 |

# 3. APPLICABILITY DOMAIN

### *3.1 Similarity-based Assessment*

**Average Similarity:** 0.289 (Threshold: 0.7)
**Status:** Outside Applicability Domain

| Training ID | SMILES | Similarity | Label |
|---|---|---|---|
| TRAIN_001 | CCO | 1.000 | Non-irritant |
| TRAIN_003 | CC(C)O | 0.200 | Non-irritant |
| TRAIN_004 | CC(=O)O | 0.182 | Irritant |
| TRAIN_005 | Oc1ccccc1 | 0.062 | Irritant |
| TRAIN_002 | c1ccccc1 | 0.000 | Irritant |

### *3.2 Leverage-based Assessment (QMRF)*

**QMRF Formula:**
$h^* = x^T (X^T X)^{-1} x$
Warning threshold: $h^* = 3p/n = 0.4200$
($n = 100$ samples, $p = 14$ features)

**Calculated Leverage:** 30.1968
**Status:** Outside Applicability Domain (WARNING)

■ **WARNING:** The compound has high leverage ($h^* > 3p/n$), indicating it is structurally different from the training set. Predictions should be used with caution.

## 4. REFERENCES & DISCLAIMER

***References:***


1. NTP Interagency Center for the Evaluation of Alternative Toxicological Methods (NICEATM)

2. EPA ECOTOX Database

3. QSAR Model Reporting Format (QMRF)


***Disclaimer:***

This prediction is generated by QSAR models and should be used for screening purposes only. The predictions are based on in vivo data and do not replace experimental testing. Users should consider the applicability domain assessment when interpreting results. Predictions outside the applicability domain should be treated with caution.